

Question	Answer	Marks
2(a)(i)	(normal) force per (unit cross-sectional) area	B1
2(a)(ii)	(due to difference in depth there is a) difference in pressure between top and bottom (of ball)	B1
	(due to pressure difference, upwards) force on bottom of ball is greater (than downwards force on top of ball, so resultant force is upwards)	B1
2(b)(i)	arrow vertically downwards labelled 'weight'	B1
	arrow vertically upwards labelled 'upthrust'	B1
	arrow vertically upwards labelled '(viscous) drag'	B1
2(b)(ii)	SI base units of D : kg m s^{-2}	C1
	base units of η : $\text{kg m s}^{-2} / (\text{m} \times \text{m s}^{-1})$ $= \text{kg m}^{-1} \text{s}^{-1}$	A1
2(c)(i)	$\text{upthrust} = 920 \times (4 / 3) \times \pi \times (4.2 \times 10^{-3})^3 \times 9.81 = 2.8 \times 10^{-3} \text{ (N)}$	A1
2(c)(ii)	weight = drag + upthrust	C1
	$(2.4 \times 10^{-3} \times 9.81) = (2.8 \times 10^{-3}) + (6\pi \times 4.7 \times 4.2 \times 10^{-3} \times v)$	C1
	$v = 0.056 \text{ m s}^{-1}$	A1

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